



How to Train Your AI Dragon: Email #4 - Data Wrangling

Hello! Welcome to email #4 of this short series for social scientists. (Was this forwarded? [Sign up here](#)).

Today's topic is near and dear to my heart: data wrangling! In this email, I cover three distinct tasks: cleaning and organizing, exploring and summarizing, and communicating visually. AI is useful at each stage, but in different ways.

To extend our dragon metaphor: Toothless can fly, carry weight, and cover ground Hiccup couldn't cover alone — but our hero, Hiccup, still has to know where they're going.

Data cleaning and organization

AI is excellent at turning unstructured lists into structured tables: names, affiliations, dates, categories. This works for participant rosters, reference lists, interview transcripts with speaker labels, survey exports with inconsistent formatting, and so on.

As a concrete example, you can paste the raw export from a survey tool (Qualtrics, REDCap) and ask AI to identify inconsistent response categories, flag likely data entry errors, and suggest a cleaning protocol. Not a replacement for cleaning the data yourself — but an excellent first pass that tells you where to look.

As for one of the most painful but important tasks, codebook generation: you can give your AI tool a dataset with variable names and ask it to draft a codebook. Then you can edit it. This is much faster than starting from scratch and is especially useful when inheriting someone else's data.

Exploratory analysis and code generation

AI dragons are generally pretty good at quickly generating well marked-up, logical code (such as R, Python, or Stata) for standard analyses. Descriptive tables, regression models, survival analysis, multilevel models -- you describe what you want; it drafts the code; you run it, check it, and modify it.

This use case is really helpful for researchers who use statistical software occasionally rather than daily, teams working across different software environments, and anyone who needs to adapt code from a methods paper to their own dataset structure. For example, if I'm working with colleagues who have SPSS code, but not Stata or R, the AI tool can rapidly translate between programming languages.

Another useful AI capability is error interpretation: paste in an error message you don't understand and ask AI to explain it and suggest a fix. This is one of the most underrated uses — it turns cryptic output into plain English without requiring you to excavate Stack Overflow.

Visualization

AI tools can generate ggplot2, matplotlib, or Tableau-ready code from a description of what you want to show. Describe the variables, the story you want to tell, and the audience, and you'll get a working draft chart in the right format to iterate from.

Another handy skill is to ask your favorite AI, "What's the right chart type for comparing three continuous variables across four ethnic groups over time?" and you'll get a useful starting point for the decision — usually with caveats about your audience and publication format. Treat this as a thinking prompt, not an instruction.

What AI cannot do and why this matters

- It cannot make analytical decisions for you. Which covariates to include in a model, whether a result is clinically meaningful, how to handle a systematic missing data pattern — these require your domain expertise and your knowledge of study design. AI will give you an answer if you ask; however, the answer may be totally off-base.
- It cannot access your data directly (unless you're using a tool with file upload, in which case: check your institution's data governance policy before uploading anything identifiable). Most useful data work with AI involves describing your data structure and pasting summary outputs — not sharing raw data.
- AI is worse with numbers than with words. It can misread tables, hallucinate statistics, and present incorrect results with the same confident fluency as correct ones. Any number AI gives you needs to be independently verified against your actual output.

Toothless gets Hiccup off the ground and across terrain that would take days on foot. But Hiccup reads the map, sets the destination, and knows when to land.

The data is yours, the interpretation is yours. AI provides the wings.

Sincerely,

Lisa M. Lines, PhD, MPH

LisaLines.com



Lisa Lines Consulting

Christchurch, New Zealand

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